

THE EVIDENCE FOR VEGANISM

Scientific Evidence on Animals, Health, Agriculture, and the Environment

A collection of findings from peer-reviewed research, scientific institutions, and major international datasets examining vegan diets and the broader impacts of animal agriculture.

This library covers human health, nutrition, land use, water, climate, biodiversity, food production, economics, public health, animal cognition, sentience, welfare, fishing, and aquaculture.

Each statement is kept concise and paired with its source and URL so the evidence can be independently examined.

The purpose is simple:

Present the evidence clearly and let the data speak for itself.

1. LAND, CROPS & FOOD EFFICIENCY

Agriculture: ~50% of Earth's habitable land

Source: FAO data / Our World in Data

URL: <https://ourworldindata.org/land-use>

Meat + dairy: ~83% farmland | ~18% calories | ~37% protein

Source: Poore & Nemecek, Science (2018)

URL: <https://doi.org/10.1126/science.aag0216>

No animal products: ~76% less agricultural land

Source: Poore & Nemecek, Science (2018)

URL: <https://doi.org/10.1126/science.aag0216>

Global soy: ~77% animal feed | ~7% human food | ~16% industry

Source: FCRN / USDA PSD via Our World in Data

Source: Animal Equality summary of FCRN / USDA data

URL: <https://ourworldindata.org/drivers-of-deforestation>

URL: <https://animalequality.org.uk/blog/soy-and-deforestation/>

Global cereals: ~48% human food | ~41% animal feed | ~11% industrial

Source: FAO data / Our World in Data

URL: <https://ourworldindata.org/land-use-diets>

Corn: >95% of U.S. feed-grain production + use

Source: USDA Economic Research Service

URL: <https://www.ers.usda.gov/topics/crops/corn-and-other-feed-grains/>

Human-edible crops → animal feed: substantial calorie loss

Source: Cassidy et al., Environmental Research Letters (2013)

URL: <https://doi.org/10.1088/1748-9326/8/3/034015>

Direct crop consumption: more people fed from the same land

Source: Cassidy et al., Environmental Research Letters (2013)

URL: <https://doi.org/10.1088/1748-9326/8/3/034015>

Global deforestation: ~90% driven by agriculture

Source: UN FAO, Global Remote Sensing Survey

URL: <https://www.fao.org/forest-resources-assessment/remote-sensing/remote-sensing-survey/en>

Agricultural deforestation: ~50% cropland | ~38.5% grazing

Source: UN FAO, Global Remote Sensing Survey

URL: <https://www.fao.org/forest-resources-assessment/remote-sensing/remote-sensing-survey/en>

2. WATER & POLLUTION

Global freshwater use: Agriculture 72% | Industry 15% | Human 13%

Source: UN FAO

URL: <https://www.fao.org/land-water/water/water-scarcity/en/>

Farm pollution: nitrogen + phosphorus → waterways

Source: UN FAO

URL: <https://www.fao.org/land-water/water/water-pollution/en/>

Nutrient pollution: eutrophication | algal blooms | dead zones

Source: U.S. Environmental Protection Agency

URL: <https://www.epa.gov/nutrientpollution>

Animal manure: nitrogen + phosphorus | runoff → waterways

Source: U.S. Environmental Protection Agency

URL: <https://www.epa.gov/npdes/animal-feeding-operations-afos>

Livestock ammonia: contributes to harmful PM2.5

Source: Domingo et al., PNAS (2021)

URL: <https://doi.org/10.1073/pnas.2013637118>

U.S. food air pollution: animal foods cause most health damage

Source: Domingo et al., PNAS (2021)

URL: <https://doi.org/10.1073/pnas.2013637118>

3. CLIMATE & BIODIVERSITY

Food systems: ~1/3 of human-caused GHG emissions

Source: Crippa et al., Nature Food (2021)

URL: <https://doi.org/10.1038/s43016-021-00225-9>

Food GHGs: ~57% animal foods | ~29% plant foods

Source: Xu et al., Nature Food (2021)

URL: <https://doi.org/10.1038/s43016-021-00358-x>

Beef: ~25% of food-production GHGs

Source: Xu et al., Nature Food (2021)

URL: <https://doi.org/10.1038/s43016-021-00358-x>

No animal products: ~49% lower GHGs | ~50% less acidification | ~49% less eutrophication

Source: Poore & Nemecek, Science (2018)

URL: <https://doi.org/10.1126/science.aaq0216>

Agricultural expansion: habitat destruction + fragmentation

Source: IPBES Global Assessment

URL: <https://www.ipbes.net/global-assessment>

50% animal-food replacement: ~31% lower farm GHGs | near-halt net forest loss by 2050

Source: Kozicka et al., Nature Communications (2023)

URL: <https://doi.org/10.1038/s41467-023-40899-2>

4. HUMAN HEALTH & NUTRITION

Well-planned vegan diets: nutritionally adequate for adults

Source: Academy of Nutrition and Dietetics (2025)

URL: <https://doi.org/10.1016/j.jand.2025.02.002>

Vegan diets: can support long-term cardiometabolic health

Source: Academy of Nutrition and Dietetics (2025)

URL: <https://doi.org/10.1016/j.jand.2025.02.002>

Protein

Plant foods: all essential amino acids | vegan diets can meet protein needs

Source: Mariotti & Gardner, Nutrients (2019)

URL: <https://doi.org/10.3390/nu11112661>

Protein combining every meal: unnecessary

Source: Mariotti & Gardner, Nutrients (2019)

URL: <https://doi.org/10.3390/nu11112661>

Fiber

Fiber: plant foods ✓ | meat 0g | eggs 0g | milk 0g

Source: NIH MedlinePlus / USDA FoodData Central

URL: <https://medlineplus.gov/dietaryfiber.html>

URL: <https://fdc.nal.usda.gov/>

Cholesterol

Whole plant foods: essentially cholesterol-free

Source: USDA FoodData Central

URL: <https://fdc.nal.usda.gov/>

Animal foods: can contain dietary cholesterol

Source: USDA FoodData Central

URL: <https://fdc.nal.usda.gov/>

Randomized Vegan Trial

Vegan RCT, 8 weeks: LDL -14 mg/dL | weight -1.9 kg | fasting insulin -2.9 µIU/mL

Source: Landry et al., JAMA Network Open (2023)

URL: <https://doi.org/10.1001/jamanetworkopen.2023.44457>

Plant Protein & Longevity

Plant protein: associated with ~8% lower mortality | ~12% lower CVD mortality

Source: Naghshi et al., BMJ (2020)

URL: <https://doi.org/10.1136/bmj.m2412>

+3% energy from plant protein: associated with ~5% lower mortality

Source: Naghshi et al., BMJ (2020)

URL: <https://doi.org/10.1136/bmj.m2412>

Plant-Based Dietary Patterns

Plant-based diets: associated with ~16% lower all-cause mortality

Source: Quek et al., Nutrition Journal (2023)

URL: <https://doi.org/10.1186/s12937-023-00877-2>

Healthy plant-based diets: ~15% lower CVD | ~21% lower Type 2 diabetes

Source: Quek et al., Nutrition Journal (2023)

URL: <https://doi.org/10.1186/s12937-023-00877-2>

Cancer

Processed meat: Group 1 carcinogen | 50g/day → ~18% higher colorectal cancer risk

Source: WHO / International Agency for Research on Cancer

URL:

<https://www.who.int/news-room/questions-and-answers/item/cancer-carcinogenicity-of-the-consumption-of-red-meat-and-processed-meat>

Red meat: Group 2A probable carcinogen

Source: WHO / International Agency for Research on Cancer

URL:

<https://www.who.int/news-room/questions-and-answers/item/cancer-carcinogenicity-of-the-consumption-of-red-meat-and-processed-meat>

Group 1: evidence strength | does not mean equal danger to tobacco

Source: WHO / International Agency for Research on Cancer

URL:

<https://www.who.int/news-room/questions-and-answers/item/cancer-carcinogenicity-of-the-consumption-of-red-meat-and-processed-meat>

Type 2 Diabetes

Higher red-meat intake: associated with higher Type 2 diabetes risk

Source: Gu et al., American Journal of Clinical Nutrition (2023)

URL: <https://doi.org/10.1016/j.ajcnut.2023.08.021>

Modeled Global Health Effects

Global vegan scenario: modeled ~8.1M fewer deaths/year by 2050

Source: Springmann et al., PNAS (2016)

URL: <https://doi.org/10.1073/pnas.1523119113>

5. PUBLIC HEALTH

Antibiotic misuse + overuse: accelerates antimicrobial resistance

Source: World Health Organization

URL: <https://www.who.int/news-room/fact-sheets/detail/antimicrobial-resistance>

Livestock antibiotics: can select for resistant microbes

Source: World Organisation for Animal Health

URL: <https://www.woah.org/en/what-we-do/global-initiatives/antimicrobial-resistance/>

Resistant microbes: animals ↔ humans ↔ ecosystems

Source: World Health Organization

URL: <https://www.who.int/news-room/fact-sheets/detail/antimicrobial-resistance>

Emerging infectious diseases: majority in major study originated in animals

Source: Jones et al., Nature (2008)

URL: <https://doi.org/10.1038/nature06536>

Livestock intensification: can affect zoonotic-disease risk

Source: Gilbert et al., Animal (2021)

URL: <https://doi.org/10.1016/j.animal.2020.100123>

6. SENTIENCE & COGNITION

Sentience: capacity to experience feelings

Source: Lambert et al., Animals (2022)

URL: <https://doi.org/10.3390/ani12091182>

Fish sentience: 349 papers | 142 species + subspecies reviewed

Source: Lambert et al., Animals (2022)

URL: <https://doi.org/10.3390/ani12091182>

Fish: pain | fear | anxiety | learning | memory

Source: Lambert et al., Animals (2022)

Source: Brown, Animal Cognition (2015)

URL: <https://doi.org/10.3390/ani12091182>

URL: <https://doi.org/10.1007/s10071-014-0761-0>

Cephalopods: octopuses | squid | cuttlefish → sentience evidence

Source: Birch et al., LSE Scientific Evidence Review (2021)

URL:

<https://www.lse.ac.uk/business/consulting/reports/review-of-the-evidence-of-sentiences-in-cephalopod-molluscs-and-decapod-crustaceans>

Decapods: crabs | lobsters | crayfish → sentience evidence

Source: Birch et al., LSE Scientific Evidence Review (2021)

URL:

<https://www.lse.ac.uk/business/consulting/reports/review-of-the-evidence-of-sentiences-in-cephalopod-molluscs-and-decapod-crustaceans>

Bivalves: nervous systems | nociception evidence | learning evidence

Source: Yurchenko et al., Frontiers in Zoology (2018)

Source: Sneddon, Journal of Experimental Biology (2015)

Source: Wageningen University & Research (2022)

URL: <https://pmc.ncbi.nlm.nih.gov/articles/PMC5896133/>

URL: <https://journals.biologists.com/jeb/article/218/7/967/14518/Pain-in-aquatic-animals>

URL: <https://research.wur.nl/en/publications/welfare-of-bivalves-in-aquaculture-a-first-literature-scan-and-pl/>

Clams + scallops: predator detection + escape behavior

Source: Crook & Walters, ILAR Journal (2011)

URL: <https://doi.org/10.1093/ilar.52.2.185>

Pigs: learning | memory | complex social behavior

Source: Marino & Colvin (2015)

URL: <https://escholarship.org/uc/item/8sx4s79c>

Chickens: learning | memory | social cognition

Source: Marino, Animal Cognition (2017)

URL: <https://doi.org/10.1007/s10071-016-1064-4>

Cattle: learning | memory | emotions | social relationships

Source: Marino & Allen (2017)

URL: <https://doi.org/10.26451/abc.04.01.06.2017>

7. ANIMAL WELFARE & FARMING

Farm animals: pain | fear | distress

Source: European Food Safety Authority

URL: <https://www.efsa.europa.eu/en/topics/topic/animal-welfare>

Animal transport: heat stress | thirst | fatigue

Source: European Food Safety Authority

URL: <https://www.efsa.europa.eu/en/topics/topic/animal-welfare>

Some farming systems: restrict highly motivated natural behaviors

Source: European Food Safety Authority

URL: <https://www.efsa.europa.eu/en/topics/topic/animal-welfare>

Commercial farming: can involve painful procedures

Source: European Food Safety Authority

URL: <https://www.efsa.europa.eu/en/topics/topic/animal-welfare>

8. FISHING & AQUACULTURE

Bycatch: unintended non-target animals | injury + death

Source: NOAA Fisheries

URL: <https://www.fisheries.noaa.gov/national/bycatch/understanding-bycatch>

Bycatch: turtles | marine mammals | seabirds | non-target fish

Source: NOAA Fisheries

URL: <https://www.fisheries.noaa.gov/national/bycatch/understanding-bycatch>

Aquaculture feed: wild fish | fishmeal | fish oil

Source: Global aquaculture-feed review

URL: <https://pmc.ncbi.nlm.nih.gov/articles/PMC11482329/>

Feed fisheries: can produce bycatch

Source: Global aquaculture-feed review

URL: <https://pmc.ncbi.nlm.nih.gov/articles/PMC11482329/>

9. ECONOMICS & RESOURCE USE

Plant-based diets: can cost less in high-income countries

Source: Springmann et al., Lancet Planetary Health (2021)

URL: [https://doi.org/10.1016/S2542-5196\(21\)00251-5](https://doi.org/10.1016/S2542-5196(21)00251-5)

Vegan diets: among cheapest modeled healthy diets

Source: Springmann et al., Lancet Planetary Health (2021)

URL: [https://doi.org/10.1016/S2542-5196\(21\)00251-5](https://doi.org/10.1016/S2542-5196(21)00251-5)

Global vegan scenario: modeled ~\$1.07T/year health savings by 2050

Source: Springmann et al., PNAS (2016)

URL: <https://doi.org/10.1073/pnas.1523119113>

Food-related climate damage: measurable economic costs

Source: Errickson et al., Nature Food (2021)

URL: <https://doi.org/10.1038/s43016-021-00265-1>

10. EVIDENCE KEY

~ = approximately

Associated = observational relationship, not automatic proof of causation

Modeled / scenario = estimated outcome under specified assumptions

RCT = randomized controlled trial

GHG = greenhouse gas

PM2.5 = fine particulate air pollution

CVD = cardiovascular disease

Group 1 carcinogen = sufficient evidence that an agent can cause cancer; it does not mean all Group 1 agents carry equal risk

EVIDENCE STANDARD

No claim enters this collection simply because it supports veganism.

Every factual claim has a source directly underneath it.

Every URL must lead to evidence supporting the claim.

Each fact appears once in the most appropriate category.

Related statistics are combined when they describe the same subject and can remain short without losing scientific meaning.

Complete distributions are kept on one line whenever the evidence supports the categories.

Additional supporting sources belong underneath the existing claim rather than creating a duplicate claim.

Numbers remain attached to their original population, comparison, timeframe, or model.

Associations remain associations.

Models remain models.

Claims stay short without changing their meaning.

Primary research, systematic reviews, major datasets, and authoritative scientific institutions are preferred.

Stronger evidence is used when available.

Unsupported claims are excluded.

Present the evidence clearly and let the data speak for itself.